

Exp No: 13

EMBEDDED C PROGRAM TO ADD AN ARRAY OF 16-BIT NUMBERS AND STORE THE 32-BIT RESULT IN INTERNAL RAM

AIM:

To write an Embedded C program to add an array of 16-bit numbers and store the results in internal RAM using Keil software.

PROGRAM:

```
#include <reg51.h>

void main(void)
{
    unsigned int i;
    unsigned int array[5] = {0x1111, 0x2222, 0x8888, 0x4444, 0xABCD};
    unsigned long sum = 0;

    for (i = 0; i < 5; i++)
    {
        sum = sum + array[i];
    }

    P0 = (unsigned char)(sum & 0xFF); // Lower 8 bits
    P1 = (unsigned char)((sum >> 8) & 0xFF); // Next 8 bits
    P2 = (unsigned char)((sum >> 16) & 0xFF); // Upper 8 bits

    while(1); // keep output stable
}
```

